



Rental Management System

1. Introduction

Problem Statement

Design and develop an enhanced Rental Management experience that enables rental businesses to efficiently monitor their operations from a single interface while automating common rental workflows. The solution should improve operational efficiency, reduce manual intervention, and provide better visibility into rental activities throughout the complete rental lifecycle.

2. Business Challenges

Rental companies commonly face the following challenges:

- No centralized dashboard to monitor ongoing rental operations.
 - Difficulty tracking products scheduled for pickup and return.
 - Manual calculation of late return charges.
 - Lack of visibility into overdue rentals requiring immediate attention.
 - Security deposits are often managed outside the rental workflow, making reconciliation difficult.
 - Limited operational insights for rental managers to prioritize daily activities.
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3. Application Workflow

Frontend

The end user/ portal user can login to the Rental website, browse the product they want to rent for and select the product and the rental period, add product to the cart.

Add the shipping details or select to collect the product from the store.

Add payment related information and make the payment with the security deposit.

After the payment, the user can download the invoice from the portal.

For return the user needs to visit the store on the specified time to return the product.

IF the user returns the product on time he will get the entire security deposit back.

If the user fails to return the product on time, the penalty will be calculated and deducted from the security deposit amount.

On portal users can access and manage all their rental order, their shopping address, profile & profile image and payment related information.

Backend

Here the Admin can configure all the organisation specific rental requirements

- Creating products & Pricelist
- Creating rental period
- Quotation Template for faster Quotation Creation

When the user comes to rent the product offline/in store, the admin will create the quotation and if the client wants to rent it on the spot, the admin will confirm the quotations and create the invoice and collect the payment with the Security Deposit.

At the time of return, the admin will check the product and timing and if everything looks good they will return the security deposit amount to the customer.

If the product is late return, the penalty will be calculated and deducted from the Security Deposit and the remaining amount refunded to the customer.

User Role

Admin

Admin has responsibility to manage organization-wide Rental Management and Customer information & Rental product records.

Responsibilities

- Create
 - Product
 - Pricelist
 - Rental Period
 - Manage the User Records
 - Configuration organization-specific Rental settings.
 - Maintain pricelist, Late fees, Deposit amount, pickup and return of the Rental Products.
 - Create Quotation Template and Header & Footer for Quotation to send to the client.
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4. Client/ Portal User

Users can log in to the portal and perform action on the website. Here users can access & manage all their orders, update their address and also update their photos.

User Can:

- Register and manage their profile
 - Browsing the product on rental website
 - Select the product and rent it for the specific period
 - Add delivery address
 - Select to collect the product from store
 - Provide payment information
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5. Authentication

The application starts with a splash screen followed by employee authentication.

Features

- Portal user Login
- New User Registration
- Profile Creation

After successful authentication, users are redirected to the dashboard.

Screens

- Splash Screen
- Login
- Sign Up

Admin has access to the system backend where they can manage all the system configuration related work and routine Rental related tasks.

1. Rental Operations Dashboard

Provide a comprehensive dashboard that offers real-time visibility into rental activities.

Possible insights include:

- Active Rentals
- Rentals Due Today
- Upcoming Pickups
- Upcoming Returns
- Overdue Rentals
- Revenue from Rentals
- Security Deposits Held
- Late Fee Collection

The dashboard should help rental managers quickly identify priorities and take appropriate actions.

2. Rental Security Deposit Management

Improve how rental deposits are managed throughout the rental lifecycle.

Possible considerations include:

- Collect security deposits during confirmation.
 - Support fixed amount or percentage-based deposits.
 - Track deposit payment status.
 - Hold deposits until products are successfully returned.
 - Once the product is successfully delivered on specified time, Security amount is refunded back to the customer without any deduction
 - If the product is returned late, the penalty would be calculated and deducted from Security Deposit
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- Maintain complete deposit history.

3. Late Return Fee Management

Automate the handling of overdue rentals.

Possible capabilities:

- Automatically detect overdue returns.
 - If the product is returned after the specified time period it would be counted as Late Return and a penalty should be applied.
 - This penalty amount would be deducted from the Security Deposit and rest amount will be refund to client in cash
- Configurable charging rules.
- Hourly, daily, weekly, or monthly late fee calculation.
- Grace period configuration.

- Maximum late fee limits.
 - Automatic invoice generation.
 - Clear visibility of outstanding penalties.
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4. Pickup & Return Management

Provide a streamlined workflow for product pickup and return.

Possible features:

Pickup

- Daily pickup schedule.
- Route or sequence planning.
- Pickup confirmation.
- Customer notifications.
- Barcode or QR code scanning.
- Pickup checklist.

Return

Daily return schedule.

- Product condition inspection.
- Damage reporting.
- Missing accessories verification.
- Return confirmation.

- Automatic stock updates.
 - Deposit settlement.
 - Late fee calculation.
 - Repair workflow initiation when required.
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5. Price & Attributes

There would be one default price list which will be applicable to all the products by default.

Possible features:

- User can create multiple pricelist as required
- Some pricelist are for specific time period
- Create product Variants like Brand, Manufacturer, Color, Size

Expected Outcome

- The proposed solution should:
- Simplify rental operations.
- Reduce manual work.
- Improve operational visibility.
- Automate repetitive rental tasks.
- Enhance customer experience.
- Enable businesses to make faster operational decisions through real-time insights.

6. Bonus Ideas

Participants are encouraged to go beyond the core requirements by introducing innovative capabilities such as:

- Predictive maintenance suggestions
 - Smart pickup route optimization
 - Automatic customer reminders
 - Product availability forecasting
 - Mobile-first rental operations
 - Barcode/QR scanning
 - IoT-enabled asset tracking
 - Customizable dashboard widgets
 - KPI and business analytics
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What Participants Will Learn

- Design and develop a real-world enterprise application.
- Build intuitive UI/UX and end-to-end business workflows.
- Implement secure authentication and role-based access control.
- Develop real-time features, dashboards, and analytical reports.
- Design scalable system architecture and business logic.
- Gain hands-on experience in full-stack application development.
- Strengthen problem-solving, collaboration, and software engineering practices.

Mockup:-

<https://app.excalidraw.com/l/65VNwvy7c4X/5l50ctoqUXw>